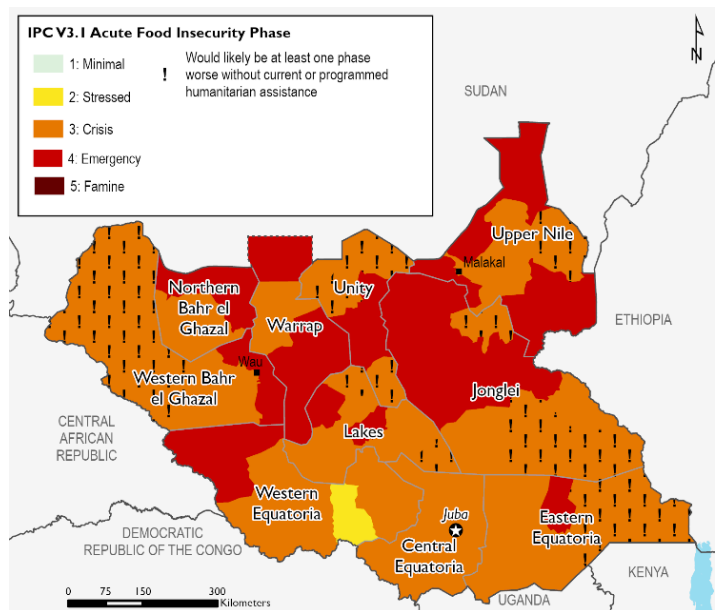


A fourth year of flooding and continued threat of conflict drive high, unmet need in South Sudan

KEY MESSAGES

- Widespread Crisis (IPC Phase 3) and Emergency (IPC Phase 4) outcomes exist across South Sudan driven by prolonged conflict and three consecutive years of flooding. Despite the delivery of humanitarian food assistance, which is improving food security in 11 counties, over a third of counties are in Emergency (IPC Phase 4). The most severe acute food insecurity exists in Unity, Jonglei, Upper Nile, Northern Bahr el Ghazal, and parts of Lakes and Eastern Equatoria.
- Despite a below-average start of the rainfall season, South Sudan is forecast to face its fourth year of atypically high flooding in 2022. The anticipated impacts of this flooding, which include crop and livestock losses as well as trade disruptions, come on top of severe flooding in 2019, 2020, and 2021. Although lingering floodwaters from the 2021 floods [substantially receded in late May](#), higher water levels do still continue to hamper household movement, trade flows, crop production, and the delivery of humanitarian food assistance, and they signal the likelihood of further flooding with above-average rainfall in later in the 2022 rainy season.

Current food security outcomes, June 2022



Source: FEWS NET

FEWS NET classification is IPC-compatible. [IPC-compatible](#) analysis follows key IPC protocols but does not necessarily reflect the consensus of national food security partners.

- Food security is expected to deteriorate between June and September, overlapping with the peak of the 2022 lean season, due to the impacts of conflict and flooding, as well as global economic supply chain disruptions, which will in turn drive high staple food prices and limited income-earning opportunities. FEWS NET estimates 7-8 million people, over half of the country's population, will need humanitarian food assistance during these months. Food security is anticipated to improve marginally during the October to January harvesting period, but many households are still expected to face consumption deficits or engage in negative coping to reduce the size of their consumption gaps.
- The critical role of humanitarian food assistance as a stopgap for acutely food insecure households is low due to a litany of financial, security, and logistic challenges, including but not limited to the rising costs of procuring and delivering food commodities due to the global impacts of the Russian-Ukraine war. Available distribution plans suggest WFP will reach less than a quarter of the national population during the lean season and only seven percent in late 2022.
- Although the occurrence of Famine (IPC Phase 5) is not considered the most likely outcome, there is a credible, alternative scenario in which Famine (IPC Phase 5) could occur, given the high proportion of the population already facing acute food insecurity and the potential for the shocks expected in South Sudan to be more severe in nature than already anticipated. Fangak and Canal/Pigi are among the areas of highest concern for this [risk of Famine \(IPC Phase 5\)](#), based on the severity of current food insecurity, the local population's very high vulnerability to new shocks, and the likely exposure to severe floods and/or renewed conflict in 2022. If flood severity exceeds 2021 and if conflict were to occur at unforeseen, extremely high levels, and if such events were to isolate households for a prolonged time, then Famine (IPC Phase 5) would likely occur.

High levels of acute malnutrition persist in South Sudan amid a continued risk of Famine

Prolonged conflict in South Sudan has remained the primary driver of the country's widespread acute food insecurity by causing large-scale population displacement, loss of able-bodied household members, limited access to agricultural fields, trade flow disruptions, and the destruction of productive assets, including livestock. Over the past three years, the effects of conflict have been compounded by the long-term impacts of the floods in 2019, 2022, and 2021, and South Sudan is forecast to face its fourth year of atypically high flooding in 2022.

These shocks and their secondary impacts are driving low food availability and disrupted food access across the country. With minimal access to crops and livestock, and amid very high staple food prices, many households rely heavily on fish, wild foods,

and humanitarian food assistance as their only food sources. However, the ability of households to expand their collection of fish and wild foods to mitigate their large food consumption gaps is limited, based on the number of people relying on these sources, persistent barriers to population movement, and the time intensity of these activities. Meanwhile, funding shortages and logistic constraints have limited the ability of humanitarians to reach more than a quarter of the population in need in South Sudan. Amid these shocks and the large-scale, unmet level of need, high levels of acute malnutrition (Figure 1) and hunger-related mortality persist in South Sudan.

Evidence suggests these shocks will only continue in the latter half of 2022. As a result, over 60 percent of the population of South Sudan will need humanitarian food assistance during the July-August peak of the lean season, and FEWS NET expects the population in need will decline only marginally with the harvest in the last quarter of 2022. In both periods, many counties will see over one in five households experience large consumption gaps indicative of Emergency (IPC Phase 4), signaling that high levels of acute malnutrition and hunger-related mortality will most likely continue. Furthermore, some households in various locations across the country will likely be in Catastrophe (IPC Phase 5)¹ throughout the outlook period.

South Sudan will continue to face one of the world's worst food security emergencies as these recurrent shocks have driven a reduction in food availability, regularly cut off household access to food, and limited households' capacity to recover from past shocks. While the occurrence of Famine (IPC Phase 5) in the projection period is not considered the most likely outcome, there is a credible, alternative scenario in which Famine (IPC Phase 5) could occur, given the high proportion of the population already facing acute food insecurity and the potential for these shocks to be more severe than already anticipated. Fangak and Canal/Pigi are among the areas of highest concern for this [risk of Famine \(IPC Phase 5\)](#), based on the severity of current food insecurity, the local population's very high vulnerability to new shocks, and the likely exposure to severe floods and/or renewed conflict in 2022. However, the risk exists to some degree across the country. If flood severity exceeds 2021 or if conflict were to occur at unforeseen levels, and if such events were to restrict household movement and isolate households in inaccessible areas where humanitarians were unable to reach the worst-affected for a prolonged period, Famine (IPC Phase 5) would likely occur. Even without the occurrence of Famine (IPC Phase 5), though, it is critical to emphasize that Emergency (IPC Phase 4) reflects an already elevated level of hunger-related mortality. A significant scale-up of food assistance is needed urgently and throughout the projection period in South Sudan to save lives.

Figure 1. Malnutrition results in South Sudan from SMART surveys conducted in 2022

County	Date	GAM (WHZ)	SAM (WHZ)
Fangak	March 2022	17.5% (13.8%-21.8%)	3.2% (2.0%-5.2%)
Canal/Pigi	March 2022	18.1% (14.9%-21.8%)	3.2% (2.0%-5.2%)
Fashoda	March 2022	15.2% (11.9%-19.3%)	3.8% (2.5%-5.6%)
Juba PoC	March 2022	19.0% (15.2%-23.6%)	4.9% (3.6%-6.7%)
Bentiu PoC	April 2022	17.6% (14.1%-21.7%)	3.3% (1.9%-5.7%)
Budi	June 2022	18.0% (13.9%-23.1%)	3.5% (2.2%-5.5%)

Source: ACF (Fangak); IMC (Canal/Pigi); Concern (Bentiu PoC and Juba PoC); AAH (Fashoda and Budi)

¹ The IPC classifies acute food insecurity at the household level and area level. At the household level, Catastrophe (IPC Phase 5) occurs when a household group has an extreme lack of food and/or other basic needs even after full employment of coping strategies. At the area level, Famine (IPC Phase 5) occurs when at least 20 percent of the households in a given area have an extreme lack of food; the Global Acute Malnutrition prevalence, as measured by weight-for-height z-score, exceeds 30 percent; and mortality, as measured by the Crude Death Rate (CDR), is greater than 2 per 10,000 per day.

NATIONAL OVERVIEW

Current Situation

Conflict: Conflict and insecurity remain the most significant drivers of acute food insecurity in South Sudan despite the recent establishment of the joint command structure and the recommitment of the Revitalized Government of National Unity to graduate the long-overdue Unified Forces (NUF). Of highest concern is **Unity State**, where an escalation of conflict has forced the displacement of 40,000 people internally and 7,000 people to neighboring Fangak of Jonglei, as well as disrupted market functioning and forced a suspension of assistance delivery. Tensions are still high following the Twic Dinka attack on the Unity State government convoy, which left five dead, mostly among Bul Nuer civilians, signaling the likelihood of revenge attacks in central Unity. Tensions have also increased between Nuer communities from Unity and Dinka communities from Pariang in the Ruweng Administrative Area due to the relocation of IDPs to the disputed Roiak area near the Panakuach oil fields of Mayom.

Eastern and Central Equatoria are of increasing concern, too, due to rising conflict between Dinka Bor cattle keepers and Equatorian farmers. Ongoing cattle raiding and violence in Equatoria are threatening first-season crop prospects and trade flows, and although some Dinka cattle keepers are returning to Jonglei, concern exists for further violence between these groups or revenge attacks against Equatorians in Jonglei.

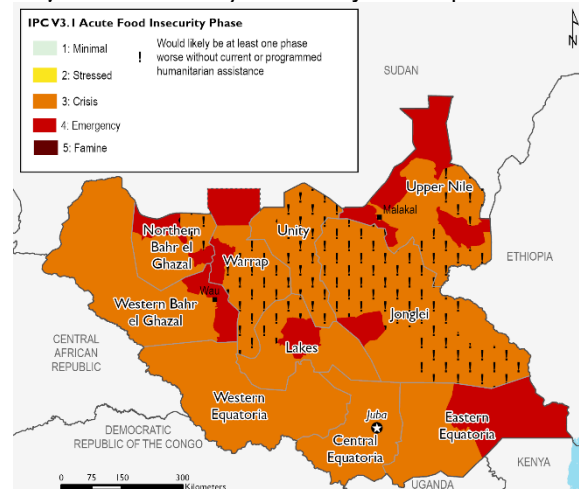
Tensions are also rising in Lankien and Pultruk of Nyirol in **Jonglei**, linked to the presence of Sudan People's Liberation Army-in-Opposition (SPLA-IO) Kitgwang Forces and the potential for clashes with SPLA-IO forces allied to Riek Machar. Similarly, tensions are high between the Gawaar Nuer from Ayod and the Lou Nuer in Nyirol and Uror. Ongoing cattle-raiding linked to the Murle has also heightened insecurity and stoked the potential for Dinka and Nuer attacks in the Greater Pibor Administrative Area (GPAA). On May 26, there was a suspected Murle raid in Duk Pajut and attacks on Nuer civilians in Akobo.

Additionally, in **Upper Nile**, armed clashes between White Army forces and pastoralists and the South Sudan People's Defence Forces (SSPDF) in Nasir have caused displacement, including forcing humanitarian workers to relocate. In **Warrap**, violence between the Ngok Dinka of Abyei and Twic Dinka from Twic has continued, notwithstanding the cessation of hostilities agreement reached in April. This was exemplified by an attack on Twic Dinka on May 22 in Mayen-Abun, which disrupted typical household movement and trade flows.

Flooding: Severe flooding in 2021 affected an estimated 835,000 people in 33 counties across eight states, with Jonglei, Unity, and Upper Nile states being the worst affected. Although floodwaters have substantially receded in late May in South Sudan, higher water levels do still continue to hamper household movement, trade flows, crop production, and the delivery of humanitarian food assistance. According to remote sensing imagery and key informants, high floodwaters have atypically persisted through the dry season, notably in Fangak and Canal/Pigi counties of Jonglei State, Melut of Upper Nile State, and Rubkona, Guit, Mayendit, and Panyijiar of Unity State. This has limited mobility, reduced cropland for cultivation, increased livestock disease, disrupted normal migration routes, and driven higher prices on markets given disrupted trade flows.

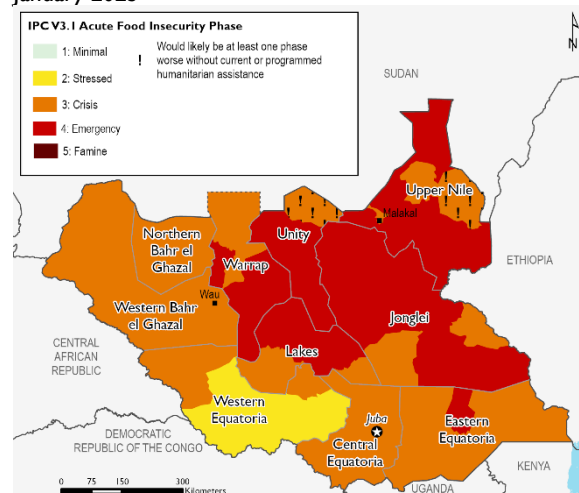
In Fangak and Canal/Pigi specifically, floodwaters have reduced since April, though some households in flooded areas still face severe movement limitations, isolating them from key food and income sources. Floodwaters in Mayendit, Panyijiar,

Projected food security outcomes, June to September 2022



Source: FEWS NET

Projected food security outcomes, October 2022 to January 2023



Source: FEWS NET

FEWS NET classification is IPC-compatible. IPC-compatible analysis follows key IPC protocols but does not necessarily reflect the consensus of national food security partners.

and Rubkona have largely not receded since April and remain a major challenge, as the standing water has limited households' access to food and income, disrupted trade flows and market functioning, and driven poor health and sanitation conditions in displacement sites. Nearly 220,000 people in flood-affected areas in Central Unity (Leer, Mayendit, and Koch) are living near flooded areas, and about 2,000 people from Mayendit remain displaced in the highlands.

The combination of atypically high floodwaters and conflict has severely disrupted humanitarian access. Key humanitarian logistical corridors were limited or closed in June, particularly from Rumbek to Panyijiar, Mayendit, and Bentiu of Unity. Two key south-north routes from Malakal and Maiwut counties to Paloich and Melut town in Melut County are also closed. These closures, alongside financial constraints, have limited the pre-positioning of life-saving food and non-food supplies in flood-prone areas prior to the peak of the lean season from June to August.

Seasonal progress: Greater Equatoria has two distinct rainy seasons, between March to May and June-September, with the crop harvest occurring in June/July and October to January. The March-May 2022 rainy season in these areas was characterized as delayed, including poor temporal and spatial rainfall distribution throughout. As a result, total cumulative rainfall was 15-45 percent below average.

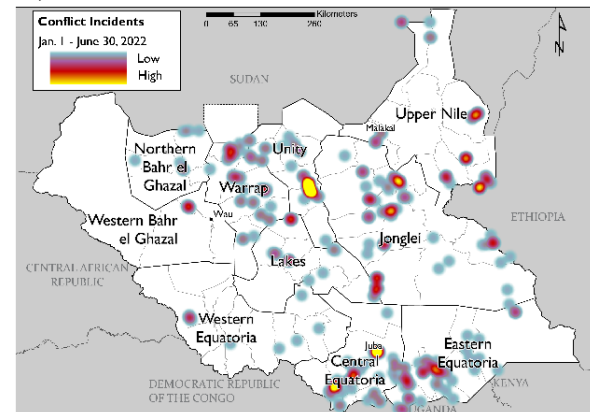
In general, the main June to September rainy season in Greater Upper Nile and Greater Bahr el Ghazal regions started on time. Meanwhile, rainfall onset was delayed in Unity and western parts of Upper Nile.

Crop production: Starting in June, households in **bimodal Greater Equatoria** typically experience an increase in food stocks and income from the first crop harvest. However, harvest prospects are poorer than normal, and first-season crop production is likely to be lower than in 2021, driven by below-average March-May 2022 rainfall, conflict, and pest infestations. According to remote sensing data, stressed soil water conditions likely reduced maize yields, including limited maize cultivation in Kit and Lokiliri *payams* of Juba County and Mugali, Ayii, Abara, and Agoro *payams* of Magwi County due to farmer-cattle herder conflict.

An assessment conducted by FEWS NET in Greater Kapoeta in May found that short-maturing sorghum and maize crops were in their seasonally typical early vegetative stages, while weeding was ongoing. In other areas, key informant and remote-sensing information suggest that while rainfall deficits negatively impacted crops in the early vegetative stages, enhanced rainfall performance between late May and early June drove improved crop conditions, and some farmers in Moggos, Nkiew, and Riwoto are re-planting the affected sorghum and maize fields. In Lopit hills of Eastern Equatoria, the early planted sorghum crops are in late vegetative stages, and in western Equatoria and parts of central and eastern Equatoria, the weeding of maize crops is nearly completed, and the crops are in the late vegetative stage. In Yambio, maize crops are at the flowering stage, and some farmers who planted maize in wetlands have started consuming maize crops.

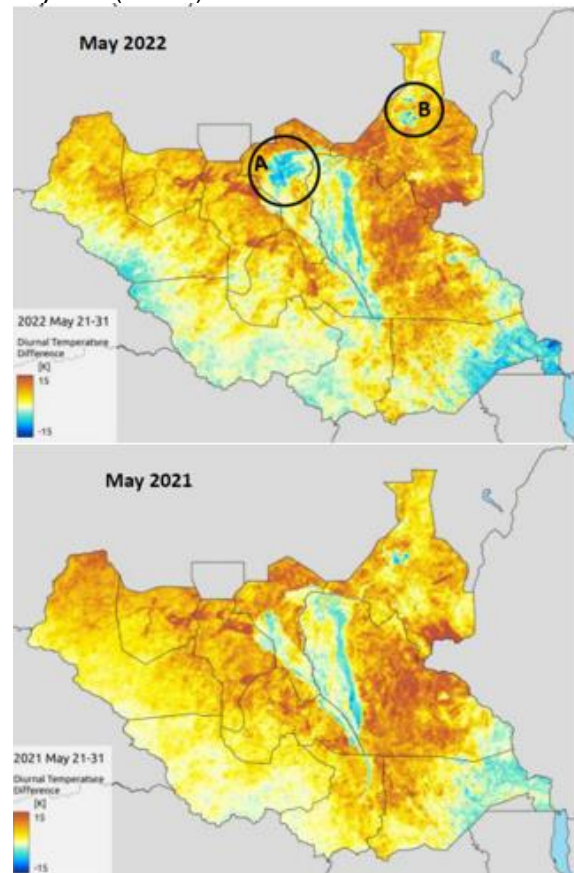
However, in Mundri West and East of Western Equatoria, field information and media reports have confirmed the infestation of maize fields by [African armyworms](#), forcing many farmers to [re-plant](#) main fields in late May. A similar report of caterpillar

Figure 2. Conflict incidents in South Sudan, January 1-June 30, 2022



Source: ACLED data

Figure 3. Flood extent as of May 2022 (top) compared to May 2021 (bottom)



Source: WFP

infestation on sorghum crops was also reported in Lofan of Eastern Equatoria.

In **unimodal Greater Upper Nile and Greater Bahr el Ghazal**, land preparation is complete in parts of the high grounds in Ayod, Fangak, Canal/Pigi, and Leer, and planting is underway in Fashoda of Upper Nile and Mayendit of Unity. However, the atypical presence of floodwaters has reportedly limited household access to farmlands and damaged some crops, particularly in Unity and parts of Jonglei and Upper Nile states (Figure 4). The outbreak of conflict in central unity and other parts of Greater Upper Nile has displaced households and limited their access to farmland. Furthermore, millipede and caterpillar infestations have been reported in much of Northern Bahr el Ghazal State and Gogrial West of Warrap State. Together these factors signal a poor progression of the season in many unimodal areas.

Livestock production: According to key informants in May, livestock in Greater Bahr Ghazal region and Upper Nile have started returning near homesteads, prompted by increased water and pasture availability. However, in many agropastoral and pastoralist areas of Jonglei, Unity, and Eastern Equatoria, conflict and the impacts of consecutive years of flooding continue to limit household access to livestock. This is on top of widespread livestock losses since 2020 when severe flooding and violence led to high livestock deaths and thefts. Where livestock have access to pasture and water, livestock conditions are reportedly fair to good in June, improving household access to milk, meat, and income from livestock products and sales. However, residual floodwaters in parts of the Greater Upper Nile region have reduced pasture availability and contaminated water sources, driving abnormal livestock migration routes. The recent outbreak of conflict, particularly in central Unity, has further restricted livestock movement to pastures, resulting in worsening body conditions. Overall, access to grazing land and water sources remains interrupted in conflict-affected (Figure 5) and flood-affected areas of the Greater Upper Nile, leading to poor livestock health conditions amidst limited availability of health services, which in turn drives livestock mortality.

In Central and Eastern Equatoria, pastoralist-farmer and intra-pastoralist conflicts continue to disrupt livestock activities. According to key informants, **increased tensions are driven by the depletion of resources**, including pasture and water, driving pastoralists from Jonglei, Lakes, and Eastern Equatoria to migrate into traditional farming areas in Greater Equatoria.

While livestock markets are functioning in most areas, low demand and poor livestock body conditions – especially in flood-affected areas – have led to lower livestock prices when compared to normal periods, reducing income from livestock products and sales. Overall, access to livestock products, including income from sales, continues to be significantly disrupted.

Macroeconomic conditions and trade: After a modest recovery from March 2021 to early January 2022, macroeconomic conditions in South Sudan have deteriorated since February 2022 due to the indirect effects of the war in Ukraine on the global economy. Global supply chain disruptions and increased commodity prices have reversed positive macroeconomic trends in 2021 through the increasing cost of food and non-food imports, depreciation of the South Sudanese Pound (SSP), and the rising cost of fuel. Even with the increased crude oil price internationally alongside South Sudan's exports of oil, an overall production decline was observed in 2022 due to the depletion of oil wells and the impact of flooding on oilfields.

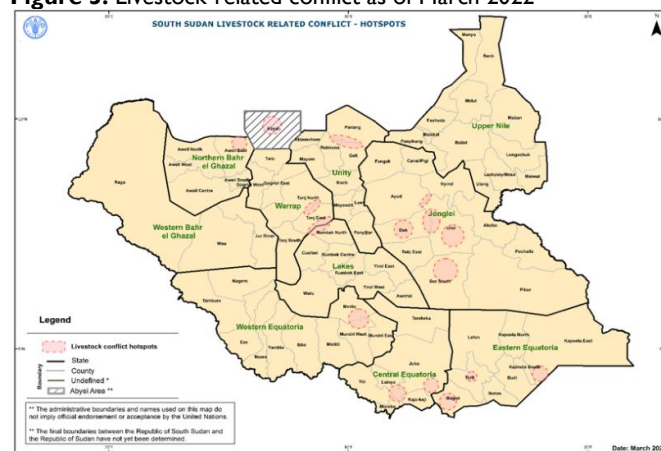
In 2021 the SSP to USD exchange rate stabilized, and the official and parallel exchange rates unified at around 400 SSP/USD. However, as global economic conditions negatively affected domestic dynamics, the SSP began depreciating in February

Figure 4. Water-logged maize crops in the early vegetative stage, July 2022, Chotbora, Fangak



Source: FEWS NET

Figure 5. Livestock-related conflict as of March 2022



Source: FAO

2022, and exchange rate trends mirrored periods of extreme exchange rate volatility, namely from 2016 to 2019. In June 2022, the exchange rate in parallel markets increased to 500 SSP/USD, with the official exchange rate increasing to 487 SSP/USD.

Despite higher import costs, food and non-food imports from neighboring countries, particularly Sudan and Uganda, are reportedly stable and, in some cases, increasing. Monthly cross-border trade data from May indicates that imports from Uganda into Equatoria via Nimule and from Sudan into Northern Bahr el Ghazal via Gok-Machar increased by 18 and 38 percent, respectively, relative to April 2022. However, high transportation costs, informal taxation along routes, and increased insecurity have negatively affected trade flows between main urban markets and smaller rural markets. Furthermore, residual floodwaters have disrupted trade flows, especially in Unity, north-central Jonglei, and south-eastern Upper Nile.

In addition, increased fuel prices have increased the cost of domestic trade. According to recent market data, the price of a liter of petrol in Juba has increased by approximately 40 percent from 500 SSP to 750 SSP from April to June 2022, with a 115 percent increase compared to the same period last year. As a result, key informants have noted that high fuel prices are driving a general increase in food and non-food prices, as traders pass higher transportation costs onto consumers.

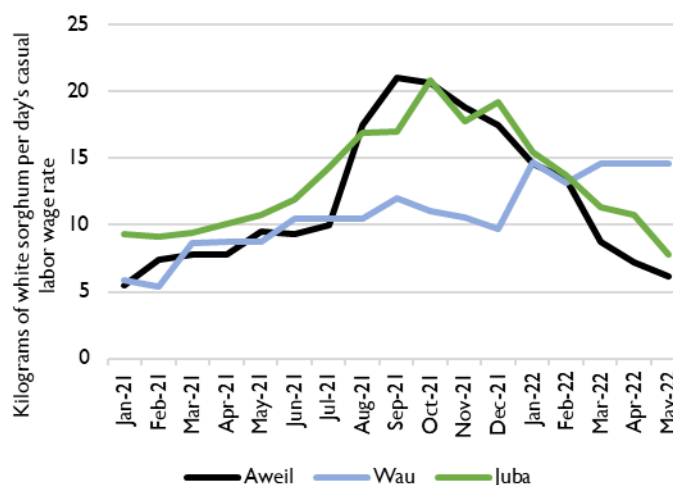
Food prices and household purchasing power: From February to June 2022, market demand has steadily increased as households deplete their food stocks from the preceding harvests. In May 2022, data from four reference markets (Rumbek Centre, Wau, Bor South, and Aweil) indicated that the retail price of a *malwa* (3.5 kg) of white sorghum was roughly 15-145 percent above April 2022 prices, driven by limited local supply, high demand, and increased transportation costs. While in Juba, the price of white sorghum was similar to April 2022, given adequate supplies. Across the five reference markets, prices were around 20-170 percent above the five-year average.

Despite the significant increase in white sorghum and fuel prices nationally, the cost of the minimum expenditure basket (MEB) in Juba in May 2022 was similar to February 2022 and increased only three percent compared to the same time last year. While MEB costs recorded in Juba remained relatively stable over the past year, key informants report that worsening macroeconomic conditions, high fuel costs, and above-average food prices have significantly limited household financial access to food, particularly for market-reliant households. In May 2022, price data showed the casual labor-to-sorghum terms of trade were lower than last year in Juba and Aweil, indicative of worsening household food access (Figure 6). However, labor-to-sorghum terms of trade were similar to last year in Wau, given the higher wage rate. With the increasing cost of living and limited income-earning options, many households limit expenditures to food and essential non-food items.

Humanitarian food assistance: Based on WFP's distribution reports, food assistance through general food distribution (GFD) and food for assets (FFA) programs reached 1.18-1.85 million people in March and April, representing an estimated 10-15 percent of the country's population. While data on actual distributions is not available for June, WFP scaled up assistance to reach 2.5 million people with GFD and FFA programs in May, about 20 percent of the national population. Populations in flood-and conflict-affected areas and IDPs remain the priorities for assistance delivery, and WFP has recently re-prioritized assistance delivery to address extreme food insecurity in Fangak, Canal/Pigi, Pibor, Cueibet, Rumbek North, Leer, Mayendit, and Tonj East, given extreme concern for acute food insecurity in these counties. Beneficiaries here are being reached with either a 21-day ration or a 30 day-ration, with deliveries occurring every 45 days. This is supported by [WFP's public reporting in June](#), which indicated 462,504 individuals in these eight counties received GFD in April.

WFP's lean season response is ongoing in counties with a large share of the population in Emergency (IPC Phase 4), including in parts of Eastern Equatoria, Jonglei, Upper Nile, Lakes, Warrap, Unity, and Northern Bahr el Ghazal. The IDP response is also ongoing in Abyei, Tambura, Aweril, and Tonj North of Warrap. WFP also resumed operations in the Mangalla IDP camp following assurances for safe food distribution for the next four months, reaching 16,000 people by early June.

Figure 6. Casual labor-to-white sorghum terms of trade in Aweil, Wau, and Juba of South Sudan, January 2021 to May 2022



Source: FEWS NET analysis of WFP data

Current outcomes

South Sudan remains one of the worst food insecurity emergencies in the world, with over 60 percent of the national population requiring humanitarian food assistance during the July-August peak of the 2022 lean season. WFP-FAO-UNICEF Food Security and Nutrition Monitoring System data from the 2021/22 harvesting period suggested that over half of the population had a poor Food Consumption Score (FCS) – marking poor dietary diversity and low food quantity – while nearly a third consumed two or fewer food groups in the 24 hours prior to the survey. Food security has likely further deteriorated since that time, given the seasonal decrease in food access as harvests deplete and as food and non-food prices have increased. Food security indicators data collected during SMART surveys in March in Budi and in June in Fashoda reported that 24-39 percent of households had poor FCS, while 24-25 percent had low HDDS. Across South Sudan, many households are facing moderate to large food consumption gaps, indicative of Crisis (IPC Phase 3) or Emergency (IPC Phase 4) outcomes. Thirty-one counties are in Emergency (IPC Phase 4), mostly concentrated in Unity, Jonglei, Upper Nile, Northern Bahr el Ghazal, and parts of Lakes and Eastern Equatoria. In 11 counties in Unity, Lakes, Jonglei, parts of Warrap, Western Bahr el Ghazal and Upper Nile, food assistance is preventing worse outcomes and Crisis! (IPC Phase 3!) outcomes exist.

Areas of most extreme concern include Fangak and Canal/Pigi of Jonglei; Leer and Mayendit of Unity; Rumbek North and Cueibet of Lakes; and Uror of Jonglei, where Emergency (IPC Phase 4) outcomes persist. A subset of the population is in Catastrophe (IPC Phase 5) as a result of the impacts of the 2021 floods on crop and livestock production and conflict/insecurity on households' ability to produce their own food and access other food sources. In other areas of concern, such as Cueibet, high staple prices, limited economic activities, the long-term impacts of inter- and intra-communal violence, and low access to other basic health and nutrition services (especially in northern Cueibet) are driving food insecurity. Koch of Unity is also of increasing concern due to the impacts of floods and conflict in April, which resulted in the displacement of many households from Mayendit to Koch. A detailed analysis is found on pages 10 to 16 of this report.

Other areas of concern include Pibor in the Greater Pibor Administrative Area and Ayod of Jonglei. Many households are facing large food consumption gaps indicative of Emergency (IPC Phase 4) due to the impacts of 2021 floods, inter-communal conflict, and cattle raids that resulted in low availability and limited access to food and income sources and asset erosion. Although the availability of natural food sources such as wild leaves, wild game, and livestock has somewhat increased in Pibor and household access to fish is also high in Ayod, some flood-affected and conflict-displaced households in isolated locations who have lost their livestock and have limited physical access to markets, food aid, and natural foods are worst affected and likely in Catastrophe (IPC Phase 5). These populations are expected in Lokoromach of Lekuangle; Manyabol, Nanaam, and Molokthoch of Gumuruk; and Labrab, Ashow, and severely flooded areas of Ayod West.

Elsewhere in South Sudan, Crisis (IPC Phase 3) or Stressed (IPC Phase 2) outcomes are present as households have relatively better access to crops, livestock, and markets but still face the negative impacts of multiple recurrent and protracted shocks such as conflict/insecurity, floods, and low 2021 crop production, and high and rising food and fuel prices.

Assumptions

The most likely scenario from June 2022 to January 2023 is based on the following national-level assumptions:

- The April 2022 agreement between SPLA-IG and SPLA-IO leadership to unify army command structures and forces has reduced political tension. However, the slow implementation of the remaining elements of the 2018 peace deal and the decentralized command structure of the armed forces continue to limit the Transitional Government of National Unity's (TGoNU) ability to enforce orders nationwide. This continues to manifest in the form of political power struggles at the state level, exacerbating the risk of clashes between armed factions in Unity, Upper Nile, and Jonglei states.
 - In Upper Nile, clashes have continued between forces loyal to First Vice President Riek Machar and armed youth militias allied to President Kiir. The cyclical nature of violence suggests conflict will continue in the area in June-July before reducing when the peak of the rainy season brings extensive flooding from August until November.
 - The frequency and severity of inter-communal conflict are likely to increase from current levels in north-central Jonglei (Nyirrol, Uror, Duk, Twic East) and the GPAA due to expected retaliatory attacks by Dinka-Nuer armed youth groups targeting Murle groups. The violence is expected to disrupt cropping activities, trade and market functioning, and humanitarian assistance deliveries.
 - In Warrap and Lakes states, a relative calm has persisted with lower levels of violence occurring during April-May compared to recent months. This reduction is attributed to a local peace initiative led by UNMISS in conjunction with the South Sudanese Government and supported by a significant disarmament campaign.

Incidents of inter-communal violence and cattle raiding are likely to reduce further during the May-October period due to expected movement constraints associated with the rainy season, flooding, and ongoing disarmament. However, insecurity is expected to persist in Tonj East and Tonj North, Cueibet, Rumbek North, and Rumbek East as herders return home with their livestock and clash with farmer communities.

- Clashes are also expected between pastoralists and farmers in Eastern and Central Equatoria, particularly in Magwi, Torit, Lainya, and Yei, and the violence will likely cause disruption to both first and second-season production. Armed confrontations are also expected across Equatoria as a result of continued delays to peace talks and the non-integration of armed opposition groups, most notably the National Salvation Army. Clashes with government security forces will likely disrupt trade flows, market functioning, and production.
 - Violence by security forces against civilians – including theft and banditry – will likely be higher than last year, peaking during the dry season in 2023. Further delays to the implementation of the Revitalized Peace Agreement, including graduating cadets into a unified national force and back payments of salary, will exacerbate raids.
- Based on the NMME, WMO, C3S, and GHACOF 61 forecasts, the June to September rainy season is expected to be above average. Given current atypically high river water levels and flood extent, the above-average rainfall over upstream river catchments in southern South Sudan and Uganda will likely drive a fourth consecutive year of floods.
 - In Greater Upper Nile and Greater Equatoria, the area planted is expected to be similar to or somewhat lower than 2021 and below the five-year average. In the Greater Upper Nile region, high residual floodwaters and the anticipated fourth consecutive season of flooding are likely to disrupt access to agricultural inputs and land available for farming, particularly in Unity and north-central Jonglei. In the Greater Equatoria, continued conflict between pastoralists and farmers and the delayed onset of the second rainy season will likely disrupt farming activities. Conversely, in the Greater Bahr el Ghazal, the area planted is likely to be higher than last year and the five-year average due to relative stability.
 - Increased insecurity is likely to lead to increased displacement, particularly in central Unity, Eastern and Central equatorial, Jonglei, and parts of Upper Nile states. Further, atypical flooding is expected to limit household mobility, isolating some communities from access to essential resources and humanitarian assistance. Areas most likely to be impacted include Unity northern and southwestern Jonglei and Upper Nile, along with parts of war app and lakes. Both increased displacement and limited household mobility are expected to restrict further household access to food, income, and critical resources.
 - Livestock production is expected to be mixed, with significantly below-average production in areas impacted by flooding and conflict. This includes Unity, Jonglei, Upper Nile, and parts of Warrap, Northern Bahr el Ghazal, and Lakes, where widespread livestock losses in the previous two years, poor pasture and water availability, and increasing incidences of livestock disease are likely to drive low livestock production and value. Further, in areas impacted by increased insecurity and cattle raids, such as Unity, Jonglei, Lakes, Warrap, and Eastern Equatoria, restricted livestock movement will limit access to traditional pastures and water sources. Dry conditions in Greater Kapoeta are also expected to limit livestock production through the start of the rainfall season. However, in areas not heavily impacted by these shocks, particularly Western Bahr el Ghazal and Lakes, livestock production is expected to improve seasonally from June to October.
 - In the Greater Upper Nile and parts of Eastern and Central Equatoria, seasonal improvement in access to fish and wild foods is expected to be below average, given increased insecurity and anticipated flooding. However, in the Greater Bahr el Ghazal and Western Equatoria, households' access will likely be average.
 - According to the World Bank, South Sudan's economy is expected to contract by 0.8 percent over the 2021-2022 fiscal year, linked largely to the continued disruption of global supply chains. As a result, food import and transportation costs are expected to remain elevated. Further, domestic oil production capacity is expected to remain limited, given the negative impact of flooding and conflict, limiting government revenue and driving further depreciation of the SSP. The SSP is expected to depreciate to or above 500 SSP/USD, driving a general increase in the price of most food and non-food items and reducing household purchasing power.
 - Imports from Uganda and Sudan are expected to decline during the second quarter, despite increased imports during the first quarter of 2022. Reduced trade flows will be driven by increased demand in source markets, lower food stock availability, increased transportation costs, worsening road conditions, and increased insecurity along cross-border trade routes. Imports from Uganda and Sudan are expected to increase during the third quarter of 2022, as the harvest improves food availability in source markets and improved road conditions increase trade flow during the dry season.

- Household reliance on markets to meet basic food needs is expected to be above average throughout the outlook period, particularly in the Greater Upper Nile region and eastern and central equatorial parts. Atypically high market reliance is expected to be driven by below-average harvest and access to livestock products for own consumption, despite a seasonal period of reduced reliance on markets.
- Based on FEWS NET's integrated price projections, which include an analysis of drivers such as local production and import levels, the retail price of white sorghum is expected to trend upwards of 100 percent higher than last year and between 20 and 170 percent above the five-year average in Wau, Juba, Aweil Center, and Bor South markets during the May to October projection period. In absolute terms, the highest sorghum price (1,200-1,700 SSP/3.5 kgs) is projected to be in Juba and Bor South, and the lowest price (447-912 SSP/3.5 kgs) is likely in Aweil Center.
- Based on WFP food assistance plans, humanitarian plan to reach a maximum of 28 percent of the national population between June and September and only 7 percent of the national population in the October and December harvesting period. With perfect targeting, this would equate to roughly 40 percent of the population in need between June and September and slightly over 10 percent of the population in need between October and December. Although food security improvements with the harvest typically drive a drop in assistance in the last quarter of the year, the relatively significant reduction in 2022 assistance deliveries is due to funding shortages, logistic and security challenges, and increased costs of food and import associated with the global rise in fuel prices. Humanitarians will likely prioritize areas of extreme concern, including Fangak, Canal/Pigi, Ayod, Pibor, Cueibet, Rumbek North, Leer, and Mayendit, which are expected to receive between a 15- and 21-day ration every 30 days.

Most Likely Food Security Outcomes

Food insecurity will further deteriorate between June and September during the 2022 lean season. Food insecurity will be driven by the impacts of conflict and flooding, as well as global economic supply chain disruptions, which will in turn lead to low household food stocks, high and rising staple food prices, and limited income-earning options. These shocks come on top of the longer-term erosion of household coping capacity that has resulted from years of protracted conflict. Additionally, further crop and livestock losses are anticipated due to the forecast of above-average rainfall and a fourth year of flooding in 2022. FEWS NET estimates that 7-8 million people will be in need of humanitarian food assistance during these months. Based on WFP's food assistance plan for June to September, food assistance is likely to reach around 3.2 million people monthly, approximately 28 percent of the national population, and – assuming perfect targeting – roughly 40 percent of the people in need of assistance. Analysis of these distribution plans and past trends suggests assistance will likely prevent more severe outcomes in 26 counties, supporting Crisis! (IPC Phase 3!) outcomes. However, Emergency (IPC Phase 4) outcomes are likely in 14 counties in Jonglei, Unity, Upper Nile, Warrap, and Lakes. Areas of most extreme concern include Fangak and Canal/Pigi of Jonglei; Leer and Mayendit of Unity; Rumbek North and Cueibet of Lakes; and Uror of Jonglei.

Between October 2022 and January 2023, food security is anticipated to improve marginally with the harvest, but many households are still expected to face consumption deficits or engage in negative coping to reduce the size of their consumption gaps. Despite the harvest, limited opportunities to earn income amid conflict and flooding, as well as high staple food prices, will continue to constrain households from purchasing sufficient food from markets. Opportunities to gain food and income from livestock production will also remain limited due to the same shocks. A fourth successive year of severe floods will also likely lead to further crop and livestock losses and disrupt food assistance delivery and trade flows. Emergency (IPC Phase 4) outcomes are expected in several counties in Jonglei, Unity, Upper Nile, Warrap, and Lakes. Food assistance will likely mitigate more severe outcomes in two counties (Maban and Pariang); however, overall assistance will have minimal impact on area-level outcomes amid the anticipated scale-down of assistance during this time.

South Sudan will continue to face one of the world's worst food security emergencies due to recurrent shocks that continue to drive severe acute food insecurity and limit households' capacity to recover from past shocks. While the occurrence of Famine (IPC Phase 5) in the projection period is not considered the most likely outcome, there is a credible, alternative scenario in which Famine (IPC Phase 5) could occur. If flood severity exceeds 2021 or if conflict were to occur at unforeseen levels, and if such events were to restrict household movement and isolate households in inaccessible areas where humanitarians were unable to reach the worst-affected households, Famine (IPC Phase 5) would likely occur. Fangak and Canal/Pigi are among the areas of highest concern for this [risk of Famine \(IPC Phase 5\)](#), based on the severity of current food insecurity, the population's high vulnerability to new shocks, and their likely exposure to severe floods and/or renewed conflict in 2022. While Famine (IPC Phase 5) is not the most likely scenario, high levels of hunger-related mortality are already occurring at Emergency (IPC Phase 4).

Events That Could Change the Outlook

Table 1. Possible events over the next eight months that could change the most likely scenario.

Area	Events	Impact on food security outcomes
<i>National</i>	A breakdown in the peace deal implementation, leading to more extreme, widespread conflicts and displacements.	A rise in conflict and insecurity events would displace more households and have an even more significant negative impact on crop production, trade flows, and market functioning. As a result, Emergency (IPC Phase 4) outcomes will be even more widespread, and an increase in the proportion of population in Catastrophe (IPC Phase 5) would be likely. A Risk of Famine (IPC Phase 5) would exist in areas most heavily impacted by the conflict.
<i>National</i>	A further deterioration in macro-economy and rise in staple food prices beyond a certain level, collapse of household purchasing powers, and further rise in criminality.	If the macro-economy deteriorates further than already anticipated, marked by spiking food prices that severely restrict household capacity to purchase even minimal amounts of food, acute food insecurity will worsen. This will also be driven by a likely corollary decline in economic activities and collapse of scale-scale businesses, leaving many households with even fewer income-earning opportunities or ability to purchase foods. These declines would likely outpace the positive impacts of assistance and the harvest, and Emergency (IPC Phase 4) outcomes would be more widespread.
<i>Fangak and Canal/Pigi, Jonglei</i>	The flood severity between July and September 2022 significantly exceeds 2021 levels and there is a noteworthy uptick in the scale and intensity of conflict.	Large-scale floods during the June to September rainfall season would result in significant household displacement, crop and further livestock losses, a disruption in trade flows and market functioning, and restricted access for food assistance delivery. If SPLA-IO further splits and conflict spreads to northern Jonglei, and the resulting conflict isolates households from accessing food and income sources, including assistance deliveries, for a prolonged time, Famine (IPC Phase 5) would be likely.
<i>Greater Pibor Administrative Area</i>	Conflict between the Murle of Pibor and the Dinka and Nuer remains at levels lower than is assumed, and flooding also occurs at lower than assumed levels, in line with seasonal norms.	Many households would have the ability to plant and harvest with lower conflict/insecurity levels. In addition, household access to natural food sources and physical markets would improve and be similar to or better than in 2021. Humanitarian access and food assistance delivery would also likely improve further from the current levels. Crisis (IPC Phase 3) outcomes would be expected.
<i>Greater Tonj of Warrap</i>	Breakdown in the local peace agreement and a lack of progress in disarmament exercise, leading to an escalation in inter-communal conflict/insecurity.	An escalation in inter-communal conflict and violence would displace households and significantly disrupt crop production and harvesting activities, trade flows, and market functioning. Additionally, the rise in conflict/insecurity would limit household access to natural food sources and hinder the delivery of food and non-food assistance delivery. Emergency (IPC Phase 4) outcomes, with high share of population in Catastrophe (IPC Phase 5) would be likely.

AREAS OF CONCERN

Jonglei State: Fangak and Canal/Pigi counties (SS06: Eastern plains sorghum and cattle livelihood zone; SS08: Nile basin fishing and agropastoral livelihood zone) (Figure 7)

Current Situation

Extensive flooding in 2021 and residual floodwater through the first quarter of 2022 have caused substantial disruption to livelihoods and food access, including widespread displacement and isolation of communities. At the onset of flooding in September 2021, over 50 percent and 30 percent of the populations of Fangak and Canal/Pigi, respectively, were affected, with IDPs and IDP returnees representing around 28 percent of the population of Fangak and 33 percent of the population of Canal/Pigi. Recent assessments, including a March 2022 SMART survey, highlighted how residual floodwaters negatively affected households, including through widespread damage to cropland, substantial livestock losses, minimal income generating opportunities, disrupted trade flows, and continued isolation from essential resources, services, and humanitarian assistance.

According to an analysis of satellite imagery, floodwaters receded in early May, improving access to humanitarian and essential services. Further, trade and market dynamics have also improved as reduced floodwaters have enabled increased trade flow and access to remote markets in previously isolated rural locations. In addition, the temporary recession of

floodwaters in Phom, Juaibor, Palei, Kuerjial, Wuntur, Tuongdiok, Pakhaan, Kuerkhan, Bei, and Biel of Fangak, has allowed the construction of dikes along the river to mitigate flood risk.

However, floodwater extent still remains significantly higher in May 2022 than in May 2021, and it continues to restrict free movement. Further, despite the brief window of improved accessibility, June marks the start of peak rainfall, and upcoming rainfall is likely to further limit mobility and access as flood waters increase.

Furthermore, many IDPs reportedly remain in Dinya, Pangaat, Tuongdiok, Pakan, Dor, Wicmuon, and Buom villages in New Fangak; and Patai, Paguir, Pulita and Wangchot, and Pulthep villages in Old Fangak. According to key informants, displaced households in isolated communities have minimal resilience to shocks currently. In addition, the recent outbreak of conflict in central Unity has led to the sudden displacement of over 7,000 IDPs into Fangak, adding additional need to already overstretched resources.

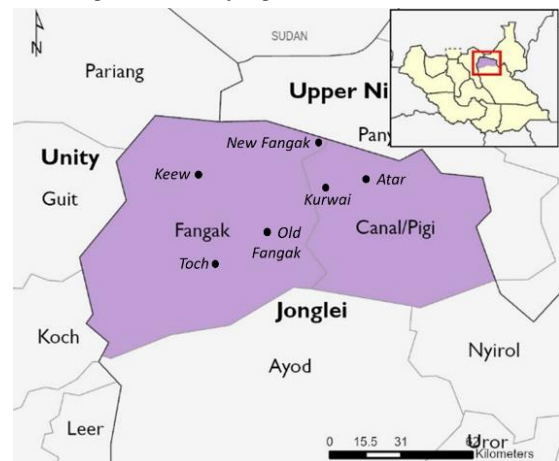
Most households in Fangak and Canal/Pigi reportedly did not plant during the previous agricultural season due to residual floodwaters, leading to a significant decrease in crop production. According to the March SMART survey, approximately 90 percent of households in Fangak and 77 percent in Canal/Pigi reported not planting crops, with approximately 70 percent of those who planted in Fangak reporting not harvesting crops. According to FEWS NET monitoring and the WFP-FAO's [Crop and Food Security Assessment Mission estimates released in June](#), cereal production declined in Fangak and Canal/Pigi by 26 and 36 percent, respectively, the lowest levels since 2013 for Fangak and the second-lowest for Canal/Pigi (Figure 8). Most households who harvested reportedly depleted their own food stocks by April, and, as a result, households have increased their reliance on fish and wild foods.

In addition to widespread crop losses, livestock ownership has significantly decreased over the last two years, showing evidence of a collapse of livelihood systems based on livestock. In a typical year, households leverage livestock to fill consumption gaps during the post-harvest period and lean season, particularly after a poor harvest. However, widespread livestock losses have resulted in severely limited access to milk, meat, and other livestock products, including income from livestock product sales. According to multiple sources, including FSNMS Round 27, the March SMART survey, key informant interviews, and interagency assessments, most households do not own livestock due to large-scale losses during the 2020 and 2021 floods. For example, the FSNMS Round 27 data indicated that only 5 percent of households in Fangak reported owning livestock, a 90 percent decline compared to 2019 (Figure 9). However, households with livestock report favorable body conditions, as most livestock have migrated to highlands where there is sufficient pasture and water availability.

In the lead-up to the main June to September rainy season, satellite-derived rainfall estimates suggest above-average rainfall in April and May. These high moisture levels are facilitating improved access to pasture for livestock but also indicate heavy rains and the potential for increased flood risk in 2022. The high soil-moisture content from residual floodwaters and lack of seeds and tools have resulted in a delay in land preparation activities. An inter-agency assessment conducted in April found no farming activities were ongoing in Old Fangak or surrounding areas, including Keew, Nyadin, and Toch. A limited number of households displaced by earlier flooding are participating in preparation activities in high-land areas or by the riverside. Some households have planted vegetables and maize on a small scale in Rier, Chotlieth, Kuerchiengjang, Wicmuon, Pakhaan, Kuerkhaan, Maskerpok, Wuntur, Tuongdiok, Bei, Biel, and Kuerthoah.

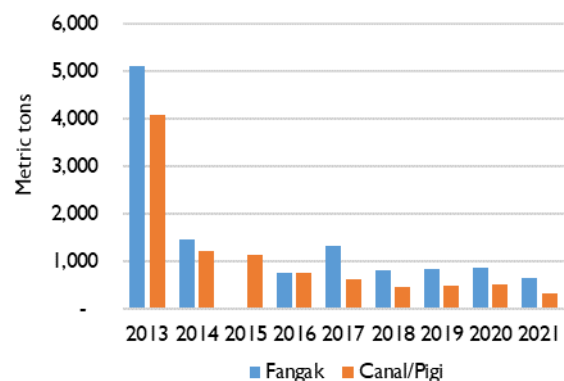
According to the March SMART survey and a recent intra-agency field assessment, nearly all households, 93 percent, still rely on fish as a primary food source, indicating a high level of access to fish but also an increased reliance on a single food

Figure 7. Area of concern reference map: Fangak and Canal/Pigi counties of Jonglei



Source: FEWS NET

Figure 8. Historical net crop production in Fangak and Canal/Pigi, 2013-2021



Source: CFSAM 2013-2021

source. In addition to increased reliance on fishing, inter-agency assessments in May reiterate a high dependence of the population on wild foods, particularly water lilies. While the receding floodwaters have led to improved access to food, as well as food assistance and markets, receding floodwaters have also reduced the quantity of available fish and wild foods.

While there has been some recovery in trade flows and market functionality, the number of traders operating in the area is low and income sources remain minimal, limited to the sale of fish, firewood, food aid, some boat hire, and remittances. Relative to February, trade routes have slightly improved in May/June with flood recession, but some localized tensions amongst and between SPLA-IO factions are interfering with the supply route from Juba and Malakal to the areas of concern.

Cereal supply is also low due to the high informal taxation imposed by multiple actors (Kitwang forces, SPLA-IO, and government) along Juba-Fangak and Malakal-Fangak and high transportation costs. Key informants have further verified that traders face challenges along riverine trade routes from Juba to Fangak, given multiple unregulated checkpoints along the Nile River. However, with the ongoing scale-up in food assistance delivery, sorghum prices have dropped from 400 SSP/kg in April to 360 SSP in May. As a result, sorghum prices are 22 percent lower than last year.

WFP continues to prioritize Fangak and Canal/Pigi, which are amongst the counties of highest concern in South Sudan. In May, WFP delivered 506 MT and 82 MT of food aid to 41,324 and 6,853 people in Fangak and Canal/Pigi, respectively. However, analysis of WFP's distribution data from March to May shows food assistance reached only 8 and 19 percent of the county populations in Canal/Pigi and Fangak. Humanitarian organizations are facing funding shortages and logistic (flooding) constraints. Although humanitarian access improved between March and May in Juaibor, Wicmuon, Dor, Manajang, Kuerkhan, Tharlier, Kuemroany, Tuongdiok, Wuntur and Phom of Fangak, and Atar 1 and Atar 2 of Canal/Pigi with the receding of floodwaters, key informants report that humanitarian organizations are still facing difficulty reaching some areas. An inter-agency assessment conducted in April confirmed that flooding and water weeds remain major challenges in accessing Kuerkang, Palei, Kuemduok, Kuerjial, Tiep, Kualdiar, Koatnyakoang, and Bei of Fangak. Moreover, food assistance delivery by speed boats or airdrops is becoming increasingly difficult and costly.

Current outcomes

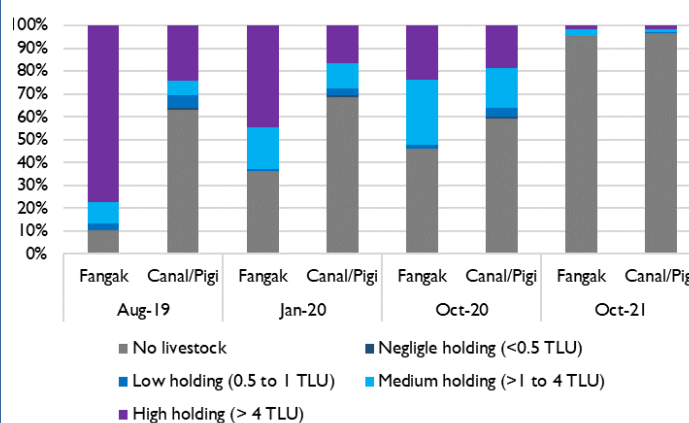
Based on food consumption data from the March 2022 SMART survey relative to the October 2021 FSNMS, the proportion of households reporting severe HHS (indicative of Catastrophe (IPC Phase 5)) declined from 46 percent in October 2021 to under 5 percent in March 2022 in Fangak while remaining somewhat steady in Canal/Pigi at 6-8 percent in both surveys. While data on food security outcomes is not available after March, the availability of fish and humanitarian assistance has likely contributed to some improvements in food security. However, contributing factors suggest households still face large consumption gaps due to a continued lack of access to livestock and no household food stocks. Furthermore, many in need do not have access to assistance, as it has reached less than 25 percent of the total need. Income-generating activities also remain minimal, and household coping capacity continues to be eroded by the repeated nature of the shocks of flooding and conflict, leaving migration to other places and begging/searching for food as some of the few coping options available to households. Emergency (IPC Phase 4) outcomes are considered likely in both counties, with households likely in Catastrophe (IPC Phase 5) in the inaccessible areas of Kurwai of Canal/Pigi; and Kuerkang, Palei, Kuemduok, Kuerjial, Tiep, Kualdiar, Koatnyakoang, Bei, Keew, Nyadin, and Toch of Fangak. Among them, the highest concern is for Keew, Nyadin, and Toch of Fangak, where humanitarians have greater difficulty accessing them due to high floodwaters and water weeds.

Assumptions

In addition to the national-level assumptions, the most likely outcomes for Fangak and Canal/Pigi counties are based on the following assumptions:

- Household engagement in cultivation is expected to be significantly reduced as atypical flooding will limit farmland availability and disrupt access to agricultural inputs. Further, both flood-driven displacement and isolation are expected

Figure 9. Household livestock ownership in 2021 compared to 2020 and 2019



Source: FSNMS

to limit households' capacity to engage in cultivation, as many IDPs are unable or unwilling to reengage in traditional agricultural activities due to fears of crop loss to floods. As a result, crop production is expected to be significantly below average for a third consecutive year.

- Household access to milk and meat from livestock for own consumption, and income from livestock products, is expected to be significantly below average, driven by negligible livestock holdings and expectations of additional livestock losses due to the fourth year of atypical flooding. Further, livestock losses from sporadic cattle raids are also expected, resulting in marginal livestock access through the outlook period.
- Similar to 2021, widespread floodwater extent will constrain trade flows along the Sudan-Malakal-northern Jonglei trade route, including increased road and river transport costs and insecurity. As a result, physical and financial trade route constraints are expected to limit the capacity of traders to restock market food and non-food supplies, maintaining high prices and negatively impacting market dynamics.
- Seasonal income-generating opportunities are expected to remain significantly below average throughout the outlook period, especially for households in locations isolated by flooding.
- Households are expected to rely heavily on natural food sources, such as wild foods and fish. The increased reliance is expected to be driven by negligible crop production and widespread livestock losses throughout 2020 and 2021, limited income-generating opportunities, and expected disruption to market supplies. However, from June to November 2022, atypical floodwater extent is expected to severely limit physical access to wild foods and fish for rural households, though IDP households in towns or displacement sites are expected to maintain some access.
- Based on WFP's humanitarian food assistance plans for July 2022 to January 2023, food assistance ration sizes of up to 21 days will likely reach at least 50 and 30 percent of the county population in Fangak and Canal/Pigi, respectively, with 55 percent of their daily kilocalorie needs on average. However, funding shortages and the forecast of above-average main season rainfall from June to September/October, with associated flooding, will likely periodically limit the delivery of planned food assistance or prevent households from traveling to distribution points. In this case, WFP is likely to implement costly airdrop operations, putting further pressure on the already scarce funding for humanitarians.

Most Likely Food Security Outcomes

Emergency (IPC Phase 4) outcomes with some households in Catastrophe (IPC Phase 5) are likely to continue from June to September in both Fangak and Canal/Pigi, overlapping with the peak of the lean season in July/August. Many households are expected to continue facing these large to extreme food consumption gaps given the lack of food stocks and extremely low or unavailable access to livestock products. Typically, household access to own food sources decreases during the lean season as household food stocks from the previous harvest are depleted, livestock migrate from the homestead to high land pastures, and households fill lean season consumption gaps through increased reliance on markets and increased consumption of wild foods and fish. However, throughout the outlook period, market dynamics are expected to remain unfavorable, with high food and non-food prices and limited income-generating opportunities.

While livestock traditionally migrate to distant highland areas during the rainy season, a common livelihood strategy is to retrieve livestock products from cattle camps for sale or consumption near the homestead. Further, typically, during periods of acute food insecurity, households may send more members to stay at the cattle camps as a coping strategy, where they can access milk and meat. However, the widespread livestock losses over the previous two years have significantly restricted household access to livestock products for consumption or income.

The current rainfall forecast for a high likelihood of flooding similar to 2021 is expected to lead to disruptions to humanitarian access, crop cultivation, household mobility, and trade routes. Households near major towns or IDP sites will likely maintain some access to fish, wild foods, and humanitarian food assistance. However, food quantity and quality will remain poor, and most households are expected to rely almost entirely on three food sources: fish, wild foods, and food assistance.

As a result, Canal/Pigi and Fangak will likely be in Emergency (IPC Phase 4) through September 2022. Households isolated by flooding that are unlikely to have access to livestock, have minimal coping capacity, and live in communities with highly strained social support structures are likely to be in Catastrophe (IPC Phase 5).

Minimal food security improvements are expected between October and January 2023, during the harvesting period, as heavy rainfall and a fourth successive year of flooding, alongside limited resources to engage in planting, will worsen harvest prospects and result in further loss of livestock. In addition, there is an increased risk of extreme food consumption gaps for households isolated by flooding and significantly reliant on humanitarian food assistance, while assistance scales down

during this period. Given these factors, Emergency (IPC Phase 4) outcomes are likely to persist. Further, livelihood coping strategies linked to atypical asset sales and income from petty trade are likely to have limited effectiveness in improving food security, given the already widespread asset ownership losses and disruption to market dynamics.

Finally, there is a credible, alternative scenario in which Famine (IPC Phase 5) could occur in Fangak and Canal/Pigi, given the large share of populations in Emergency (IPC Phase 4) or higher. Fangak and Canal/Pigi are among the areas of highest concern for a [risk of Famine \(IPC Phase 5\)](#) in South Sudan, based on the severity of current food insecurity, the local population's very high vulnerability to new shocks, and the likely exposure to severe floods and/or renewed conflict in 2022. If flood severity exceeds 2021 and if conflict were to occur at unforeseen, extremely high levels, and if such events were to isolate households for a prolonged time, then Famine (IPC Phase 5) would likely occur.

Unity State: Koch, Leer, and Mayendit counties (SS07: Nile Basin Fishing and Agropastoral livelihood zone; SS08: Northwestern Floodplain Sorghum and Cattle livelihood zone; SS09: Northwestern Nile Basin Cattle and Maize livelihood zone) (Figure 10)

Current Situation

Conflict – including extreme [violence against civilians](#) – has driven high levels of acute food insecurity in central Unity (Koch, Leer, and Mayendit counties) since early 2022, leading to large-scale population displacement, loss of able-bodied household members, and the destruction of productive assets among households who are still recovering from the 2021 floods. Violence reached a peak in April, when armed youth militias allied with the South Sudan People Defense Forces (SSPDF) clashed with counterparts allied with the Sudan People's Liberation Army-in-Opposition (SPLA-IO). Available estimates produced by an inter-agency needs assessment, the IOM Displacement Tracking and Monitoring service, and other key informants indicate around 40,000 to 53,500 people (nearly 52 percent of the population) were displaced within Leer; around 2,000 to 18,000 people (5-30 percent of the population) were displaced within Mayendit; and around 900 people (1 percent of the population) were displaced within Koch between February and May. In addition, around 7,000 people were displaced from Leer to surrounding areas, primarily Fangak County of Jonglei State.

Figure 10. Area of concern reference map: Koch, Leer, and Mayendit of Unity



In the aftermath of the politicized violence in April, households in Mayendit and Leer have been further subjected to lower levels of inter-communal conflict and cattle raids throughout [May and early June](#). These events have not only discouraged displaced households from returning to their homesteads and nearby fields and grazing areas, but also disrupted or periodically prevented households' ability to fish, gather wild fruits and vegetables, plant crops for the June-September growing season, protect their surviving [livestock](#), or access markets. Most displaced households in Leer have been unable to return home either due to violence or lingering floodwaters. Northern Mayendit has fared slightly better, with key informants reporting the return of around 11,000 people to their area of origin.

The impacts of conflict have compounded the long-term impacts of the 2021, 2020, and 2019 floods and increased the vulnerability of households to flooding in 2022, especially in worst-affected areas in Leer and Mayendit. Atypical floodwaters remained in Leer and Mayendit throughout the dry season, leaving soil saturated and inhospitable to crop growth and livestock grazing; making household, commercial, and humanitarian movements more cumbersome; and elevating disease incidence levels in displacement sites. It has also increased the chances of rapid inundation during the 2022 rainy season. Even though the 2022 rainfall season is off to a slow start, increased streamflow from upper river catchments in Uganda has already led to rising inundation in the Sudd Wetland. Based on a [UNOSAT analysis](#), nearly 220,000 people across Leer, Mayendit, and Koch are exposed to or living near flooded areas, amounting to nearly 90 percent of the combined population.

The above shocks have led to a precipitous decline in food availability and access across all three counties, but especially Leer and Mayendit. Typically, May to August is the peak of the cropping and lean season in central Unity, leaving households to depend on a mix of livestock production, fishing, wild food gathering, and market-dependent activities to collect, barter, or purchase food. However, the lean season began much earlier than usual this year. The WFP-FAO Crop and Food Security Assessment Mission (CFSAM) found that only 35 percent of households in Leer and Mayendit and 45 percent of households in Koch planted in 2021, a steep decline compared to the pre-conflict range of 72-90 percent recorded in 2013. Field

observations suggest most households that harvested had already depleted their stocks by around February/March. Furthermore, WFP-FAO Food Security and Nutrition Monitoring System (FSNMS) data indicate livestock ownership and herd sizes have sharply declined in these traditionally agropastoral areas, with around 80, 70, and 30 percent of households reporting no livestock ownership in Leer, Mayendit, and Koch, respectively, in late 2021. Among households that still own livestock, floodwaters and insecurity significantly constrain access to grazing areas, water resources, and markets, with adverse effects on livestock body condition, milk productivity, reproduction, and salability.

The low availability of income-generating activities outside of agriculture, coupled with limited to minimal market functioning and flood and conflict-related disruptions to trade and market systems, further diminish households' ability to access sufficient food. While information about markets in Koch is unavailable, key informants report low activity in local village markets in Touchriak, Thornyor, Dindin, Padeah, and Gandor of Leer, and reduced activity in the major port town of Adok of Leer as well as Rubkuay of Mayendit. Low market supply has led to a sharp increase in the retail price of red sorghum, further reducing already low household purchasing power. Based on FEWS NET market monitoring data, the retail price of a *malwa* (3.5 kg) of red sorghum in May 2022 was 20 percent higher than in May 2021 in Mayendit and 38 percent higher in Leer. The rising price of sorghum has led to a 60 percent decline in the amount of sorghum that a household can purchase with income earned from the sale of a goat in Mayendit and a 25 percent decline in Leer compared to last year. In Koch, the price of sorghum in the main Mir Mir market is likely similar to higher than Leer based on qualitative information.

As a result, most households in Leer and Mayendit and a significant share of households in Koch are atypically reliant on fishing and gathering wild foods (vegetables, fruits, water lilies) for their food needs. However, the ability of households to expand the collection of fish and wild foods to mitigate the large consumption gap left by crops and livestock production is limited, based on the number of people relying on them, persistent barriers to population movement, and the time intensity of these activities. Available [baseline information](#) collected in 2011/2012 shows that fish and wild foods typically contributed around 13-15 percent and 37-38 percent of annual kilocalorie needs in the SS08 and SS07 livelihood zones, respectively. Key informants report that many poor and displaced households who lack fishing equipment or have lost their livestock are relying only on sales of firewood and face incredible difficulty purchasing highly-priced cereals in local functional markets.

Meanwhile, given the above shocks, crop production prospects for the 2022 growing season starting in May/June are poor. Currently, the primary constraints to crop production include inundation of lowland areas, a lack of safe access to farmland, and very low access to seeds and inputs. Key informants report that land preparation and planting generally did not occur in Leer in May since most of the population is displaced in Leer Town, the Muom Military Training Centre, and surrounding areas. In dry highland areas of Leer, Mayendit, and Koch, soil moisture is supportive of cultivation despite below-average rainfall in June; however, key informants report that low access to seeds and tools – especially among the displaced – have limited sorghum and maize planting, though the International Rescue Committee (IRC) has distributed some seeds.

Finally, the critical role of humanitarian food assistance as a stopgap for acutely food insecure households has diminished due to a litany of financial, security, and logistic challenges, including but not limited to the rising costs of procuring and delivering food commodities due to the global impacts of the Russian-Ukraine war. According to WFP distribution reports for the period of March to May 2022, only an average of 2 percent of the population in Koch, 13 percent of the population in Mayendit, and 18 percent of the population in Leer received assistance monthly through the GFD or FFA programs. These figures represent 2-21 percent of the estimated population in Crisis (IPC Phase 3) or worse in each county, assuming perfect targeting. Although WFP is prioritizing counties with a large share of the population in Emergency (IPC Phase 4) or worse, including central Unity, its efforts were further complicated by the temporary evacuation of humanitarians and limits placed on humanitarian movement due to prevailing conflict/insecurity, which led to the suspension of assistance in April. Access improved in May, though, and key informants confirmed WFP and partners were able to resume operations in late May and reach at least 25,500 people in Rubkuay, Thaker, and Dablual *payams* of Mayendit, and Mayendit headquarters.

Based on past food insecurity data, the large deficit of household food and income sources, and the erosion of household coping capacity due to recent conflict and flood events, FEWS NET assesses that Emergency (IPC Phase 4) outcomes persist across Leer, Mayendit, and Koch. FSNMS R27 data collected during the 2021 harvest period found over half of the population in each county had a poor FCS and a moderate to severe HHS, with most consuming only up to two food groups. A majority of the population was already resorting to Emergency coping strategies, primarily searching and begging for food. While freedom of movement is marginally better compared to the peak of flooding in late 2021 and the peak of conflict in April/May, most households face consumption gaps, especially in Leer and Mayendit, where livestock ownership is lower than Koch. Analysis of historical and current drivers of acute malnutrition, including reduced food intake, also points to elevated, Critical (15-29.9 percent) levels of GAM. Moreover, given the high share of households in Leer and Mayendit who lack livestock and face barriers to market access, it is likely that some households without access to food assistance are in

Catastrophe (IPC Phase 5). Available information suggests the worst-affected households are located in Leah, Buath, Lual, Jak, Dhorlieth, Kuok, Luom, Bargok, Dhorjack, Gier, Tuoy, Nyal, and Dablual *payams* in Mayendit; Touchriak, Thornyor, Dindin, Padeah, and Gandor *payams* in Leer; and Bieh, Reng, Kuertuoy, Wichduelyaka, and parts of Mirmir *payams* in Koch.

Assumptions

In addition to the national-level assumptions, the most likely outcomes for Leer, Mayendit, and Koch counties are based on the following assumptions:

- The recurrent and retaliatory nature of violence and cattle raids between armed youth groups suggests violence will likely continue in July before relatively declining amid heavy rainfall and flooding between August and November. Conflict/insecurity will drive population displacement, limit household investment in crop and livestock production, and impede trade and market functioning. Past trends suggest Leer and Mayendit will be worse affected than Koch.
- Based on rising floodwaters in central Unity in June and the forecast of above-average rainfall between July and September, central Unity is at high risk of extensive floods during the growing and harvest periods. The anticipated scale of inundation will most likely cause or prolong population displacement, impede planting and cause crop damage, and contribute to low to minimal trade and market activity, especially in lowland areas in Mayendit and Leer.
- While fish and wild food availability typically increases between June and October, heavy rains and rising floodwaters simultaneously increase the dangers of fishing and gathering until floodwaters start to recede after October or November. The chances of renewed insecurity as floodwaters recede will further impede access. Access is expected to be relatively better in Koch than Leer and Mayendit due to the lower risk of flooding and insecurity.
- Based on recent trends in crop production, the limited scale of planting observed in June, anticipated levels of conflict/insecurity and flooding, and associated levels of population displacement, the 2022 harvest is expected to be lower than last in Leer and Mayendit and similar to lower than last year in Koch.
- Based on the dynamics of inter-communal cattle raids, the expected impacts of flooding on pasture availability and water contamination, and limited financial resources to restock, livestock ownership and herd sizes are likely to remain very low in Leer and Mayendit and moderate in Koch. As a result, food and income from livestock products, including milk, are expected to remain significantly below normal throughout the scenario period, regardless of seasonality.
- Based on WFP's food assistance distribution plans, at least 25 percent of the population is expected to receive a 21-day ration each month in Koch, Leer, and Mayendit, on average, from June to September. Periodic delays or suspensions of food assistance delivery are likely to occur, however, due to conflict/insecurity and flood-related logistical challenges. Levels of pre-positioned assistance in flood-prone areas are currently low due to prior funding and security constraints. Funding is reportedly inadequate to support food aid during the October-January harvest period in central Unity.

Most Likely Food Security Outcomes

From June to September, the severity and scale of acute food insecurity are expected to reach an annual peak in central Unity. Renewed flooding and recurrent conflict/insecurity are expected to restrict or altogether prevent a large share of households from engaging in crop production, while also inhibiting the recovery of livestock production and market systems and interfering with food gathering and income-generating activities. As a result, households will have unreliable and inconsistent sources of food and income, primarily consisting of fish, wild foods, and a small amount of purchased grains. Food assistance delivery will be the main factor mitigating the size of food consumption gaps at the household level, but at least 20 percent of the population in all three counties will likely have a large kilocalorie deficit, given both the scale of the population in need and the likelihood of disruptions to food assistance delivery. Furthermore, limited access to health and nutrition services, exposure to water contamination during the rainy season, and high water-borne disease incidence among children will intertwine with low food intake to drive high levels of acute malnutrition. As a result, Emergency (IPC Phase 4) outcomes are expected to persist in all three counties. Very poor, displaced households without productive assets who face physical barriers to accessing food assistance in Leer and Mayendit will likely be in Catastrophe (IPC Phase 5).

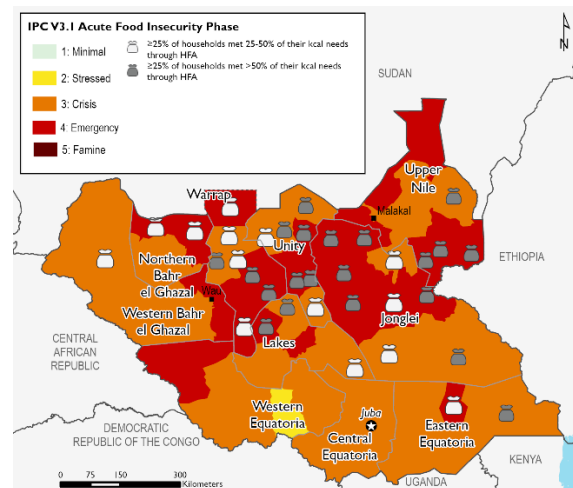
From October to January, the share of the population facing Emergency (IPC Phase 4) or worse outcomes is expected to marginally decline due to the availability of the harvest in secure, highland areas coupled with increased freedom of population movement and increased market activity as floodwaters gradually recede. However, past trends suggest the threat posed by recurrent conflict/insecurity and prolonged flood extent will likely leave tens of thousands of households displaced with minimal to no access to own-produced crops or livestock holdings. As a result, a large share of the population

in low-lying and insecure areas will remain dependent on fishing, gathering, and petty trade and labor activities, the scale of which is insufficient to prevent large food consumption gaps and/or emergency coping strategies, particularly given high cereal prices and very low purchasing power. Emergency (IPC Phase 4) outcomes are likely. Furthermore, given the lack of food assistance during this period, it is likely that some very poor households in Leer and Mayendit who have been repeatedly exposed to multiple shocks over several years, lack productive assets, and have lost their traditional coping capacity will take the risk of remaining in remote, flood-affected areas, where they can search for fish and wild leaves but are exposed to extreme hunger indicative of Catastrophe (IPC Phase 5).

MOST LIKELY FOOD SECURITY OUTCOMES AND AREAS RECEIVING SIGNIFICANT LEVELS OF HUMANITARIAN ASSISTANCE*

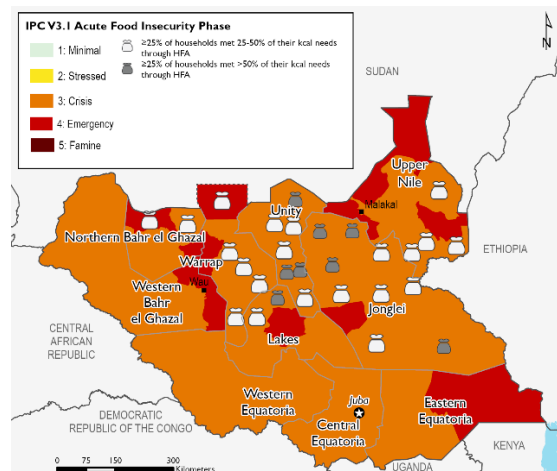
Each of these maps adheres to IPC v3.1 humanitarian assistance mapping protocols and flags where significant levels of humanitarian assistance are being/are expected to be provided. ☼ indicates that at least 25 percent of households receive on average 25–50 percent of caloric needs from humanitarian food assistance (HFA). ☼ indicates that at least 25 percent of households receive on average over 50 percent of caloric needs through HFA. This mapping protocol differs from the (!) protocol used in the maps at the top of the report. The use of (!) indicates areas that would likely be at least one phase worse in the absence of current or programmed humanitarian assistance.

Current food security outcomes, June 2022



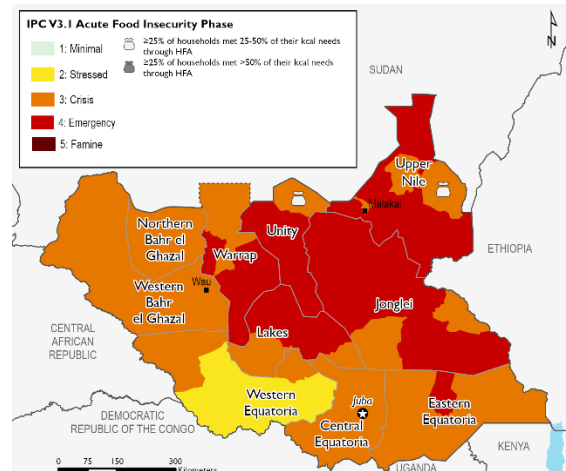
Source: FEWS NET

Projected food security outcomes, June to October 2022



Source: FEWS NET

Projected food security outcomes, October 2022 to January 2023



Source: FEWS NET

FEWS NET classification is IPC-compatible. IPC-compatible analysis follows key IPC protocols but does not necessarily reflect the consensus of national food security partners.

FEWS NET: South Sudan Food Security Outlook October 2022 to January 2023: A fourth year of flooding and continued threat of conflict drive high, unmet need in South Sudan, 2022.

ABOUT SCENARIO DEVELOPMENT

To project food security outcomes, FEWS NET develops a set of assumptions about likely events, their effects, and the probable responses of various actors. FEWS NET analyzes these assumptions in the context of current conditions and local livelihoods to arrive at a most likely scenario for the coming eight months. [Learn more here.](#)